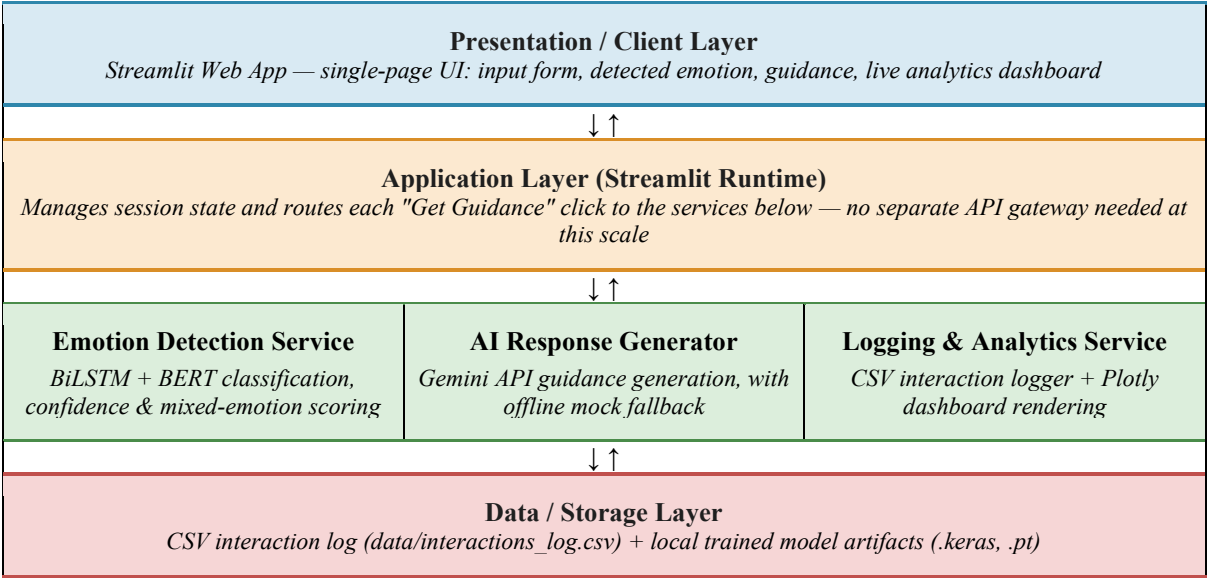


Solution Architecture

Date	12 July 2026
Team ID	
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

High-level view of the Emotion Detection & Learning Support Engine, mapped onto a simplified Presentation / Application / Data layout. As this is a single Streamlit application rather than a distributed microservice system, the middle layer represents the app's internal service modules rather than a separate API gateway.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Renders the input form, detected emotion, guidance text, and live analytics dashboard.	Streamlit
Application Layer (Streamlit Runtime)	Manages session state, handles the "Get Guidance" click, and orchestrates calls to detection, response, and logging modules.	Python 3.11, Streamlit session_state
Emotion Detection Service	Classifies free-text input into one of five emotions with confidence and mixed-emotion scores.	TensorFlow/Keras (BiLSTM), PyTorch/Transformers (BERT), NumPy
AI Response Generator	Produces personalized guidance tailored to the detected emotion; falls back to a mock handler if the API is unavailable.	Google Gemini API, Python
Logging & Analytics Service	Persists every interaction to CSV and renders emotion/confidence trend charts.	pandas, csv, Plotly

Data / Storage Layer	Stores interaction history and holds the trained model artifacts used at inference time.	CSV files, local filesystem (.keras, .pt)
----------------------	--	--